Task 4 - Sketches + Mockup

AI Emotional Companion for Elderly Users

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May 21, 2026

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1 Abstract

Our project is an AI Emotional Companion for elderly people who live alone or have limited social contact. The goal is to help them with both the emotional and the practical sides of getting old at home, since the two things are often connected and existing devices usually only cover one of them.

The system is a small smart display with a speaker that sits in a central place at home, like the living room or the kitchen. It is always on and you don't need any technical knowledge to use it. The main way to interact with it is by talking, but it also has a touchscreen for people who prefer tapping and a visible emergency button for urgent situations.

The smart display is paired with a wrist health monitor (which we added in Task 3) that provides vitals, location and presence detection. There is also a companion mobile app for a trusted family member, who gets real-time notifications and a weekly Emotional Health Report.

Apart from reacting to commands, the system is also proactive. It listens to vocal cues like tone, pace and unusual silence, together with sensor data and daily routines, to detect early signs of low mood, fatigue or a possible medical issue. It can step in with conversation, music, medication reminders or a suggested call to a relative, and it escalates to family or emergency services when vitals and certain keywords suggest something might be wrong.

This document presents five sketches that show the system and its main interaction flows, followed by a digital mockup of the device and its main screens. We also describe the technologies behind the solution, the changes we made compared to Tasks 2 and 3, and our reflection on the process.

2 Sketches

We made five sketches in total. The first one shows the overall concept of the system, and the other four show details of specific screens and flows. We kept the style low-fidelity on purpose, with simple lines, big text and very little colour, because we wanted them to feel like quick exploratory drawings instead of finished UI.

After discussing how we wanted them to look based on what we have been developing in the previous tasks, we drew the sketches digitally in Figma so the whole submission would be consistent. We asked Claude for some visual suggestions, but the actual layouts and the final drawings were done by us in Figma.

2.1 Sketch 1: Overall Concept

The first sketch shows the whole system at a glance. The smart display is in the centre, the wrist monitor is on Martha's side (left) and the family app is on James' side (right). The dashed arrows show the data flow between them, with vitals and presence going from the wrist to the device, and alerts and weekly reports going from the device to the family app. The small voice waves around the device remind the viewer that voice is the main channel.

We used this sketch as the entry point to the rest of the document, since it makes the three-part structure of the system clear before going into detail.

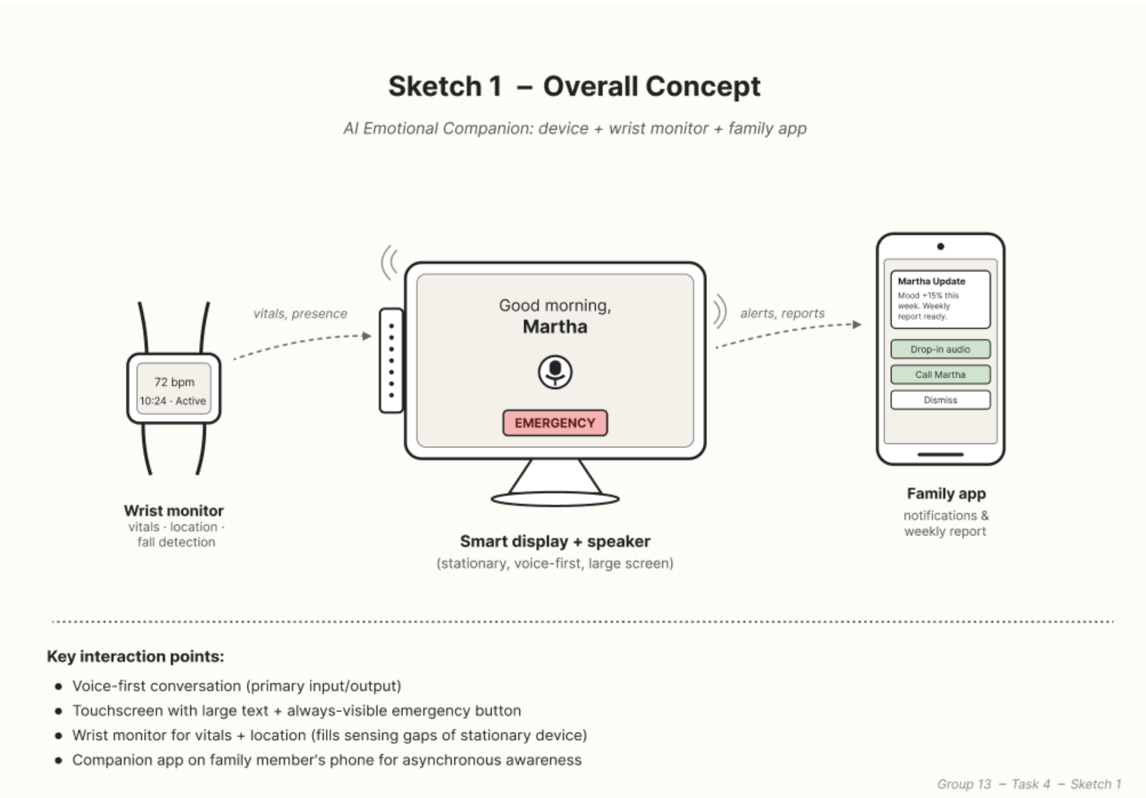

Figure 1: Sketch 1. Overall concept: device, wrist monitor and family app with the main data flows.

2.2 Sketch 2: Home Screen

The second sketch zooms into the home screen the user sees by default. We added annotations on the sides to explain each design choice. The text is large (around 22pt minimum), which is important for people with vision problems. The microphone button is big and centred because voice is the main way to interact, and the emergency button is always visible so the user doesn't have to look for it. We also limited the navigation to three items, because longer menus get overwhelming for non-technical users.

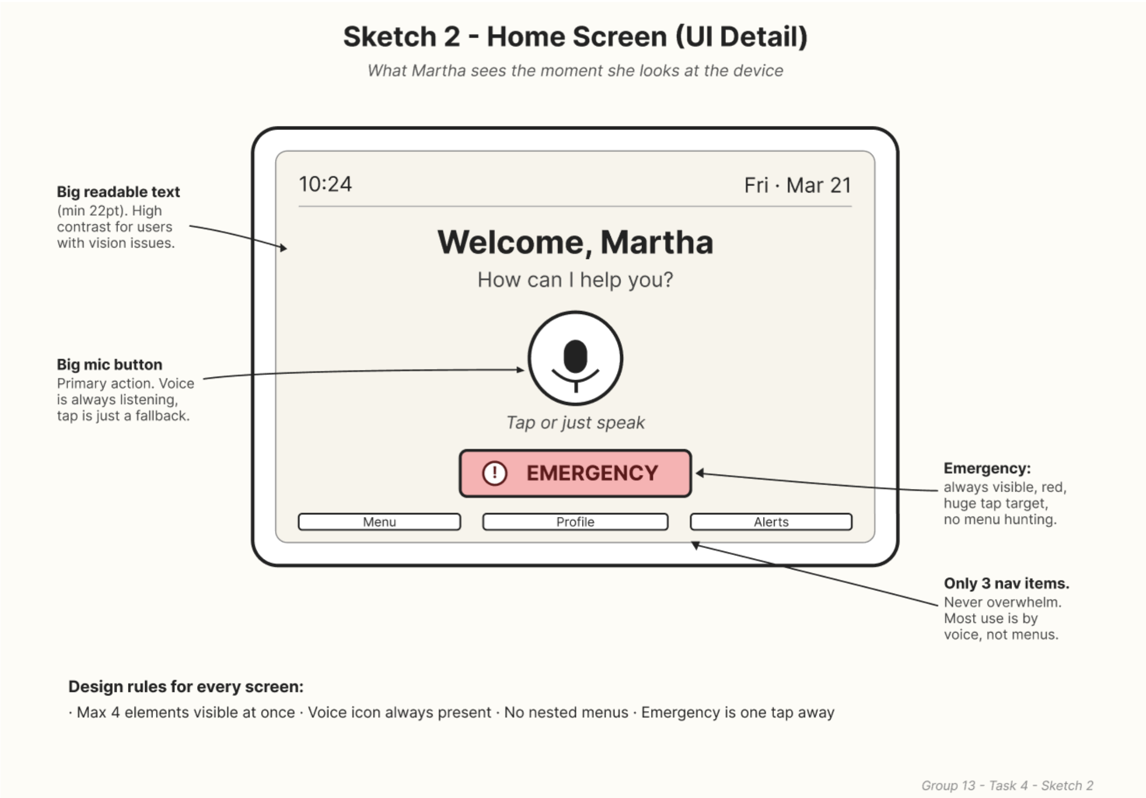

Figure 2: Sketch 2. Home screen with annotations explaining each design rule.

2.3 Sketch 3: Proactive Check-in

The third sketch shows the proactive check-in in three frames. In the first one, the system detects a long silence. In the second one, the system speaks first with a gentle question. In the third one, the system offers a clear one-tap or one-word response, in this case to call James.

Below the three frames we listed the eight situations that can trigger the system to speak (waking up, scheduled medication, long silence, evening loneliness patterns, flat tone, slow speech, medical keywords and abnormal vitals). We also added two important behaviours: if the user says "not now", the system stays silent for one hour and respects that. But if vitals and keywords suggest a possible emergency, the system speaks even if it was asked to stay quiet. This priority logic was one of the things we figured out while drawing this sketch.

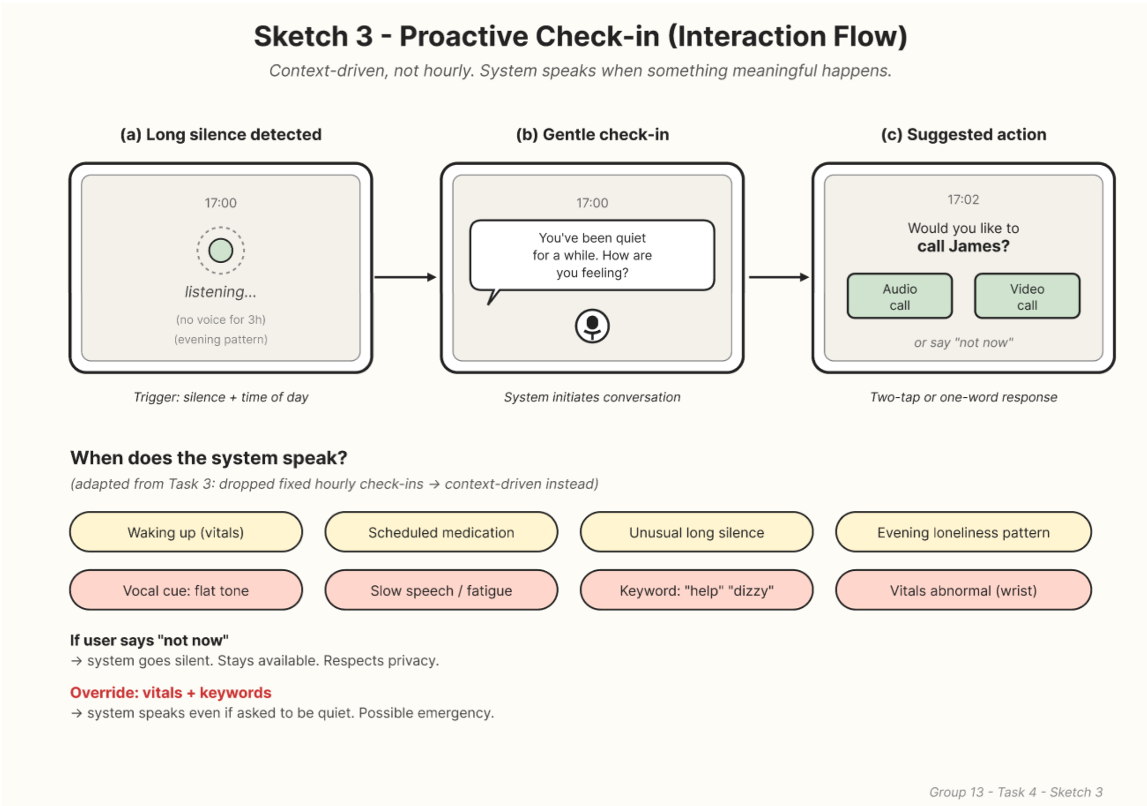

Figure 3: Sketch 3. Proactive check-in flow with the logic that decides when the system speaks.

2.4 Sketch 4: Medication Reminder Flow

The fourth sketch covers the medication reminder flow. The top row shows the happy path: the reminder appears, the user says "Done", and the system logs it. The bottom row shows what happens if the user can't find their medication. The system gives spoken guidance ("Kitchen, top cupboard, yellow sticker, blue box"), the user replies "Found, but it's empty", and the system sends a notification to James to buy a new pack.

We replaced the buttons from Task 2 with voice confirmations like "Done", "Found" and "Next" because Martha has arthritis and saying a word is easier for her than tapping a small button. It is a small detail but it matters for the user.

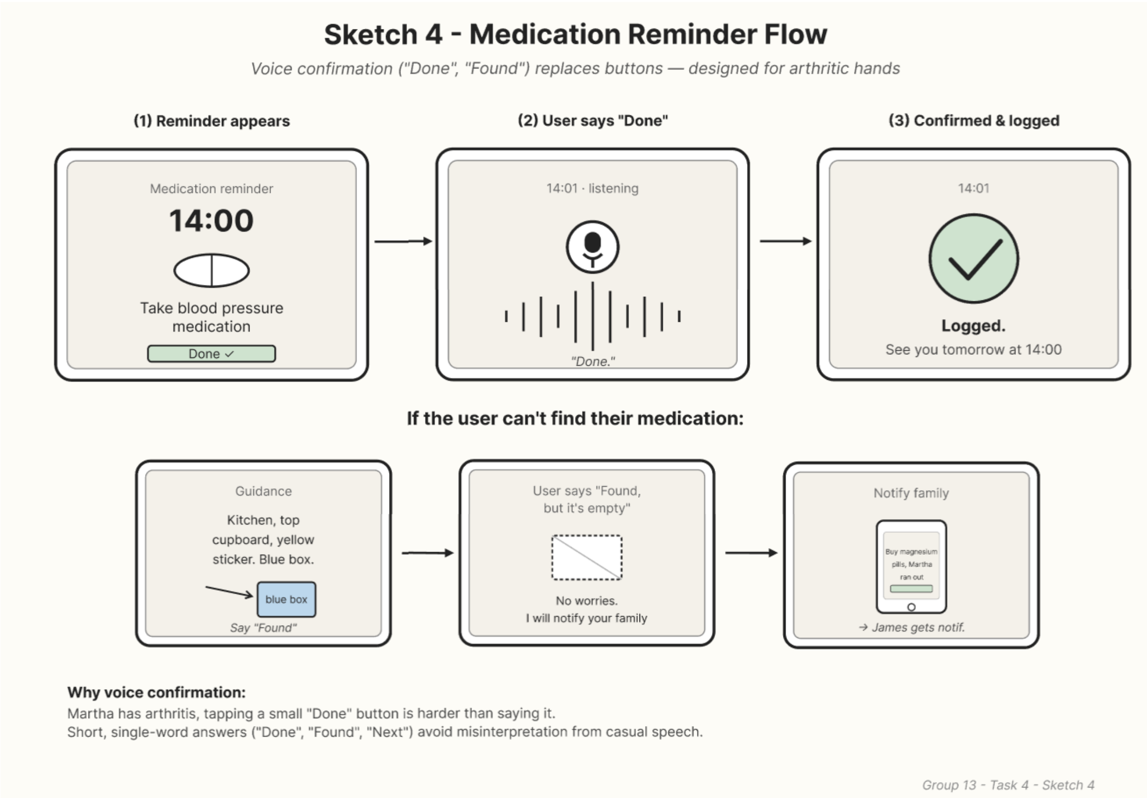

Figure 4: Sketch 4. Medication reminder with voice confirmation and a fallback when the medication has run out.

2.5 Sketch 5: Emergency Escalation Chain

The fifth sketch addresses the failure scenario from Task 3, where Martha said she felt "dizzy" and "weak" but the system thought she was sad and suggested music. The escalation chain has three steps. First, abnormal vitals are detected and the system asks loudly if she is okay. Second, if there is no response, the system sends alerts in parallel to the family and to a second trusted contact (a neighbour, in our case Melisa). Third, if there is still no answer, the system places a direct call to emergency services with the location and a snapshot of the vitals.

We also added a small decision tree to show how the three steps connect to the trigger conditions, and a panel that explains which problems from the Task 3 failure scenario this sketch is meant to fix.

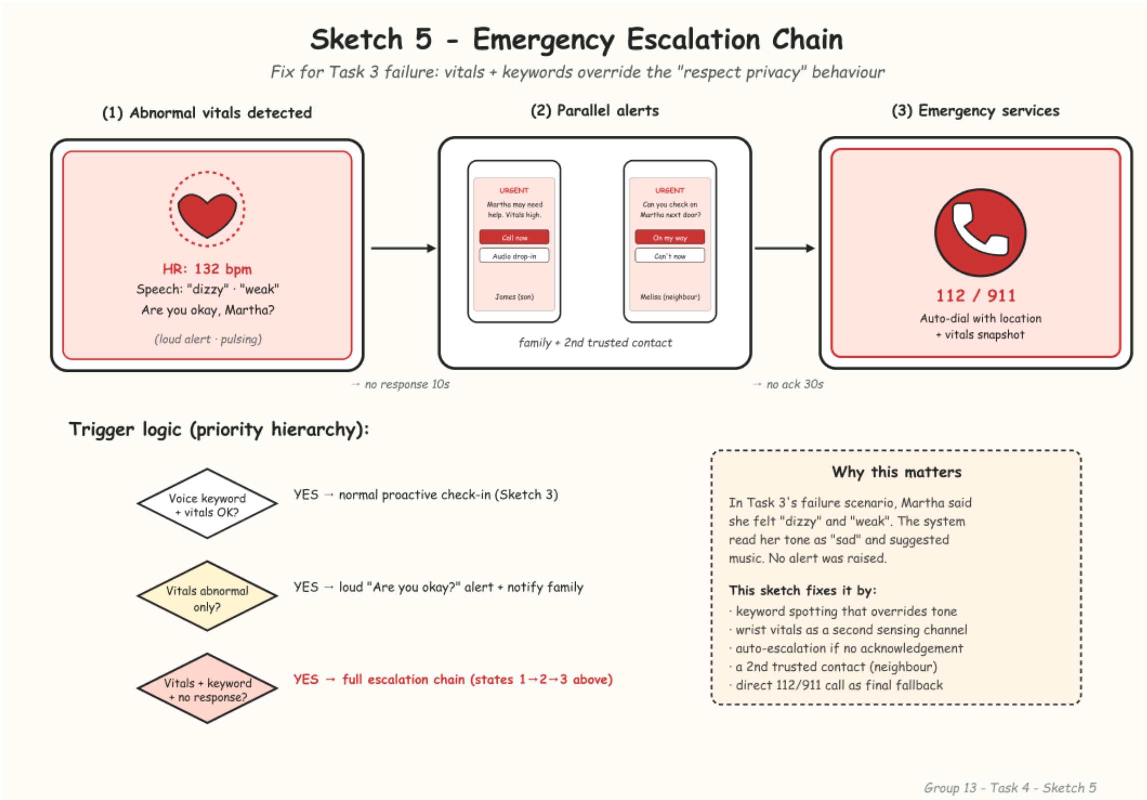

Figure 5: Sketch 5. Emergency escalation chain, made to fix the failure scenario described in Task 3.

3 Mockup

We chose to make a digital mockup instead of a physical model. The reason is that the interaction between Martha and the system is mostly about screens, speech and software behaviour. The physical form of the device is neutral and domestic on purpose, and a cardboard model would not show as much as a digital one that shows both the object and its UI states.

The mockup was made by us in Figma. We used Claude to get some visual ideas and suggestions on what details to add, but every layout, colour and screen was decided by hand in Figma so we could keep full control over the result and make sure it fits our concept.

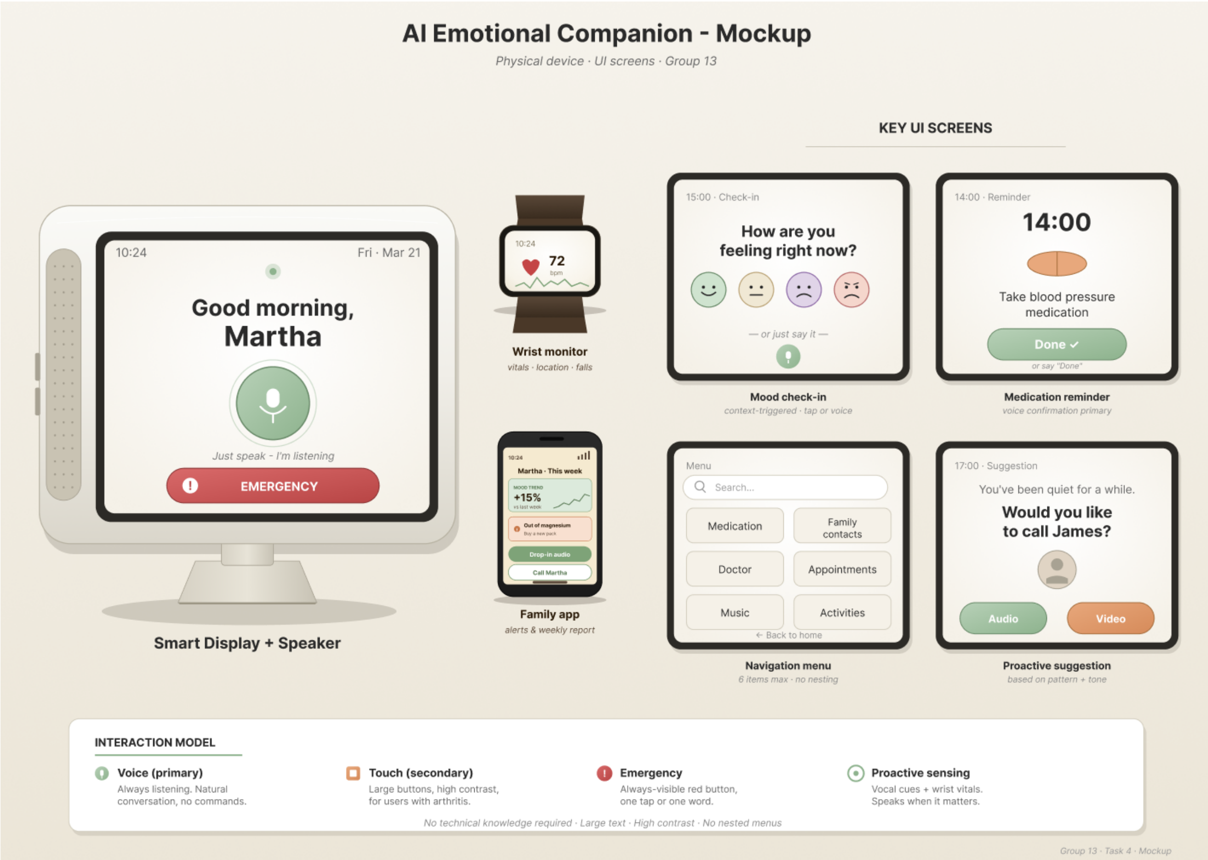

Figure 6: Digital mockup. Smart display on the left, four UI screens on the right, wrist monitor and family app below, and the interaction model at the bottom.

3.1 Central interaction points between AI and human users

The mockup shows the main AI-human interaction points:

- The microphone button on the home screen is the main interaction point. Its size and position make it clear that speaking is the expected way to use the device.
- The proactive prompt screen ("You've been quiet for a while. Call James?") shows the AI taking the initiative based on context, instead of waiting for a command.
- The mood check-in screen shows the AI asking a low-effort question, with emoji-style taps and voice as response options.
- The medication reminder shows the AI scheduling and confirming a task through a one-word voice confirmation.

- The wrist monitor (shown beside the main device) is the silent channel. Martha doesn't really interact with it, but it gives the system information she could not communicate herself, like her heart rate or whether she has fallen.
- The family app on James' phone is the asynchronous side of the system. He doesn't talk to it the way Martha talks to the device, he just gets notifications when something happens and the weekly mood report at the end of the week.

3.2 How the AI supports human activity

Each screen shows a different way the AI helps Martha:

- **Cognitive support** (medication reminder): the AI remembers and reminds, so she doesn't have to.
- **Emotional support** (proactive suggestion): the AI notices when she has been quiet and offers a way to reconnect.
- **Self-awareness support** (mood check-in): the AI invites reflection on her emotional state without being pushy.
- **Navigation support** (menu): the AI organises everything in a flat, six-item structure with no nested levels.
- **Physical safety** (wrist monitor): the watch keeps track of her vitals and location in the background, so the system can react even when she can't speak.
- **Family awareness** (family app): James doesn't have to call every day to check on her. The app keeps him updated through notifications and a weekly report, which gives Martha space and gives him peace of mind.

3.3 Why the mockup feels higher quality than the sketches

The mockup is better quality than the sketches because of a few specific things. The colour palette is consistent and limited: sage green for the AI (calm), warm orange as a secondary colour, and a muted red only for emergencies. The typography is bigger and the four UI screens are at a size you can actually read. We also added a "Companion Devices" row below the main device to show the wrist monitor and the family app in context, which makes it clearer that they are part of the same system and not separate things.

3.4 Additional Design Material

For a better-quality view of the sketches and mockups, the full design file can be accessed here:

Figma design file

4 Technologies

This section describes the technologies needed to build our solution. We split them into two groups: technologies that already exist and we would just integrate, and technologies that we think that do not really exist yet in the form we need and would require some work on our side.

4.1 Existing technologies we would integrate

Speech recognition. The device needs to turn what the user says into text the system can work with. This kind of technology is widely available now and works well even for older users or people with small speech variations. The input is the audio from the microphone and the output is the transcribed text.

Speech synthesis. The opposite direction. The system needs to talk back in a calm, natural voice. Modern voice synthesis tools sound close to a real person, which is important for us since the device is supposed to feel like a friendly presence and not a robotic assistant.

A conversational AI model. This is the part that actually understands what the user is saying and generates a response. It also needs to remember some short context, like recent interactions, the time of day, the medication schedule or the latest vitals, so the conversation feels coherent. Ideally a smaller model runs directly on the device for speed and privacy, with a bigger one in the cloud as a backup for more complex requests.

Emotion recognition from voice. A separate layer listens to how the user speaks (tone, pace, pauses) instead of just what they say. From these features it estimates if the user sounds calm, tired or low. This is what allows the system to act proactively instead of just waiting for commands.

A wrist-worn health monitor. We would not build the wrist device ourselves. There are already commercial wearables that measure heart rate, oxygen levels, falls and approximate location, and they share this data through public APIs that the system can read.

Always-on listening with privacy in mind. The device needs to know when the user is talking to it without sending all the audio to the cloud. There are small, low-power models that run on the device itself and only activate the rest of the system when someone is talking. This protects privacy and reduces how much data leaves the home.

Notifications to the family app. For James to get the alerts and the weekly report, the system needs the standard infrastructure that any mobile app uses to send push notifications. Nothing custom here.

Voice and video calls. When Martha wants to call James, the system uses standard video-calling technology to set up the call. Again, this already exists and is easy to integrate.

4.2 Technologies that do not yet fully exist

A model that combines emotional and medical signals. This is the most distinctive part of our system. We would need something that looks at three things at the same time: how the user is speaking (tone, pace), what the user is saying (specific words like "dizzy", "weak" or "help"), and the vitals from the wrist monitor. The output should be a single state: calm, low mood, tired, or possible medical emergency.

Right now, emotion recognition systems only look at voice, and health monitors only look at vitals. A combined model that tells "feeling sad" apart from "feeling physically unwell" (which is exactly what was missing in the Task 3 failure) does not exist as a ready-to-use product. To build it we would need to collect data from real users in different states and train it ourselves.

A system that knows when to speak. Most AI assistants today only respond when you talk to them. Our system needs the opposite: it has to decide on its own when it is a good moment to say something and when to stay quiet. This kind of proactive behaviour is still being researched and it is not something we could just plug in. We would start with clear rules (silence for too long, time of day, mood trend) and let the system learn over time what works best for each user.

Keeping personal data on the device. For the system to really know Martha (her favourite music, daily routine, baseline voice) it needs to learn from her over time. But sending all of that to the cloud would be a privacy issue. There are techniques that allow learning while keeping the data on the device, but they are still developing and would need some custom work.

4.3 How the technologies combine

The stack works in four layers:

1. **Sensing:** The wrist monitor sends vitals and movement data; the microphone captures the audio stream.
2. **Interpretation:** Speech recognition turns the audio into text (what is being said), and emotion recognition extracts vocal features (how it is being said). Together with the vitals, these three signals produce a unified picture of how the user is doing.
3. **Decision:** That picture feeds into the proactive behaviour module, which decides whether to speak or stay quiet. If it speaks, the conversational AI generates a response and the voice synthesis turns it into audio.
4. **Notification:** In parallel, the family app receives event-driven alerts and a weekly report that summarises the same information over time.

5 Adaptations from previous tasks

While working on the sketches and the mockup, we changed a few things from Tasks 2 and 3. Some of these changes were already announced in Task 3 (the wrist monitor and the family app), but some came up specifically during this task.

Voice confirmations instead of small buttons. Task 2 had a small "Done" tap button for the medication reminder. Once we drew Sketch 4 and remembered that Martha has arthritis, we realised a one-word answer is easier for her than tapping. We changed all the confirmation actions in our flows to follow this pattern.

Emergency escalation chain. Task 3 had a failure scenario where the system mistook a medical issue for an emotional one and never raised an alarm. In Sketch 5 we designed a proper escalation chain (loud check-in, then family plus second trusted contact, then emergency services). This is new compared to Tasks 2 and 3.

Priority logic for proactive behaviour. We made the decision logic explicit. By default the system respects the user's wish to be left alone ("not now" silences it for an hour), but vitals plus medical keywords always override that. This was implicit before, and Sketch 3 forced us to make it concrete.

Second trusted contact (neighbour). At first only James was the fallback for any alert. Sketch 5 made us realise that if James is in a meeting or sleeping, no one would respond. So we added a second trusted contact, the neighbour Melisa from the storyboards, so the system has more than one human fallback before calling emergency services.

Mood check-in with emoji and voice. In Task 2 the hourly check-in was just a voice question. In the mockup we made it a small screen with four emoji options plus a voice prompt. This makes it lower effort for the user, especially on days when she doesn't feel like talking.

Specifications of the physical device. In Task 2 we were a bit vague about the size and material of the device. For the mockup we decided around 25 cm wide, an 8" touchscreen, a single front-firing speaker and a matte ceramic finish. The reason for ceramic is that it should look like a domestic object (a vase, a teapot) and not like medical equipment.

6 Group reflection

6.1 On the process

Working on the sketches and the mockup was useful because it forced us to be specific about things we had handled abstractly before. When we tried to draw the home screen, we had to commit to what is on it and what isn't. When we drew the medication flow, we had to decide if the confirmation was a tap or a word. Small details that we could leave open in writing had to be solved on paper.

The idea changed in a few places. The biggest change was going from fixed hourly check-ins (Task 2) to context-driven ones (already in Task 3 and now confirmed in Sketch 3), because asking "How are you feeling?" every hour would become noise really fast. Another change was adding the second trusted contact for emergencies, which only became obvious once we drew the escalation chain step by step.

We used Claude as a brainstorming partner. It helped us think of visual ideas for the screens and suggest small details to add to the mockup. We did not use it to do the drawings: we made everything in Figma. This split worked well for us because Claude is fast at giving options but Figma is where we actually control how things look. AI tools were useful for the brainstorming phase and less useful for the execution phase, where having one shared canvas mattered more than having more ideas.

6.2 On the solution

Now that we see the whole solution together, we are happy with how the three parts work together: the smart display in the centre, the wrist monitor as a second sensing channel, and the family app as the asynchronous link to a caregiver. The forms of interaction also make sense for the project: voice is good for a user who finds touchscreens hard, the emergency button is a deliberate exception to the voice-first rule (because in a crisis voice might not work), and touch is just a fallback rather than the main channel.

We made some choices on purpose. We dropped the idea (which came up while brainstorming with AI) of letting the device control home appliances like the kettle, because it goes beyond our emotional and health support scope and it would add real safety risks. We kept the tone of the AI calm and gentle instead of efficient and transactional, since our user benefits more from a friendly presence than from a fast assistant. We also made the device respect privacy: when Martha says "not now", the system backs off, except when vitals clearly suggest a medical event.

We picked the materials and palette with the same idea. The matte ceramic finish and the warm colours in the mockup are there to make the device feel like part of the home, not like medical equipment. The accent colours (sage green for AI actions, warm orange for secondary, muted red only for emergency) are restrained on purpose, because an elderly user doesn't need a colourful, busy interface.

7 AI tools reflection

Tools used: Claude (for brainstorming and visual suggestions) and Figma (for the actual sketches and mockup).

We used Claude as a brainstorming partner during the design phase. We described our solution and asked it to suggest how a sketch or mockup screen could look, what details we could add to make the composition feel more finished, and how to combine the device, the wrist monitor and the family app in one image without it feeling too crowded. Claude gave us some useful starting points (like the idea of grouping the four UI states as a row at the bottom of the mockup), but it did not produce the final artifacts.

The sketches and the mockup were drawn by us in Figma. Doing them ourselves was important for two reasons. First, we wanted the final result to match our concept exactly, including small details like the voice confirmation words, the colour palette and the layout of the family app. Second, working in Figma let us keep everything editable in one place, which is what we will use for the rest of the course anyway.

There were some suggestions from Claude that we did not use. At one point it proposed adding small icons next to every UI element to make the mockup look richer, but this would have gone against our rule of keeping the screens minimal for an elderly user. It also suggested a glassmorphism effect on the screen tiles, but we did not use it because it would lower legibility and look too modern for a domestic device. These are good examples of why we kept the execution on our side: AI is fast at giving options but it doesn't always know which ones fit the project.